ZJU-UIUC INSTITUTE
Zhejiang University-University of Illinois Urbana-Champaign Institute
浙江大学伊利诺伊大学
厄巴纳香槟校区联合学院

Curriculum of Mechanical Engineering For Class of 2027

Zhejiang University/University of Illinois Urbana-Champaign Institute

(ZJU-UIUC INSTITUTE)

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1. Overview

The Zhejiang University/University of Illinois Urbana-Champaign Institute (ZJU-UIUC Institute) offers a joint dual-degree engineering program. Upon successful completion of the program, and after meeting the graduation requirements of both universities, students will obtain bachelor's degrees separately from Zhejiang University (ZJU) and the University of Illinois Urbana-Champaign (UIUC).

Mechanical engineering may be the most diverse of the engineering fields, embracing many sub-fields and affecting all aspects of our lives. Mechanical engineers work on new machines, products, and processes that hold the promise of better lives for all of us. They are concerned with both technological and economic aspects in the design, development, and use of their products. Today, one of the challenges is to design efficient, low-cost machines and processes that use the fewest possible natural resources to improve the lives of people throughout the world.

The technical portion of the Mechanical Engineering program curriculum is designed as a sequence of increasingly specialized experiences. The student's first year is spent mastering the basics of science: math, chemistry, and physics. Building on this base, in the second year, students begin to take fundamental engineering courses such as statics, dynamics, basic circuits and electronics, thermodynamics, and strength of materials. By the third year, students are taking specialized mechanical engineering courses in the sub-fields of fluid mechanics, heat transfer, dynamic systems and controls, materials, mechanical design, and manufacturing. Finally, during the senior year, students have the opportunity to both broaden and deepen their knowledge of the field through technical elective courses. At the end of the curriculum, students take the capstone senior design course in which the knowledge and skills they have learned are applied to projects submitted to the department by industrial firms or by faculty members. Engineering design, communication, teamwork, and laboratory experiences are integrated throughout the curriculum from the first year to the last year.

During their junior and senior years students are encouraged to take courses and involve themselves in research projects in cross-disciplinary areas with Civil and Environmental Engineering, Electrical Engineering, Computer Engineering, and others.

2. Graduation Requirement

2.1 Graduation Requirements for UIUC degree

2.1.1 Grade Point Average Requirement

A student must maintain a minimum GPA of 2.0 (A=4.0) to remain in good standing and graduate. UIUC degree is subjected to the final review by UIUC.

2.1.2 Junior Eligibility Requirement

To be eligible to enroll in the ME courses listed in the third year of the curriculum, a student must have completed, with a combined 2.25 grade point average, the mathematics, physics, computer science, and electrical and computer engineering courses listed in the first two years.

2.1.3 Curriculum Requirement

The curriculum leading to the degree of Bachelor of Science in Mechanical Engineering at UIUC requires **128** academic credits (or academic units, AUs) and is organized into required courses, technical electives, general education, and other electives.

i). Required courses, see section 3 for details.	Academic Credit
a) Orientation and Professional Development	1
b) Foundational Mathematics and Science	29
c) Technical Core	53
d) Composition	8
e) Advanced Composition*	4
Total Required AUs:	**95**

ii). Elective courses, see section 4 for details.	
a) Science Electives	4
b) Statistic Electives	3
c) MechSE Electives	6
d) Technical Electives	6
e) General education	12
f) Free Electives	6
Total Required AUs:	**37**

**Students take ME 470+498 to satisfy advance composition and technical elective requirement, and credits can be given for both.*

2.1.4 Additional requirements

For UIUC degree, in addition to the specific course and grade point average requirements listed above, each candidate for a bachelor's degree from UIUC must meet the following requirements:

i). Residency Requirement

Earn a minimum sixty (**60**) semester AUs of UIUC credit, of which at least twenty-one (**21**) AUs must be 300 or 400 level UIUC credit courses.

ii). Transfer Requirement

Students should have a satisfactory English Proficiency Test score on TOEFL, IELTS or other exams approved by UIUC, and maintain a good standing on academic studies that all term GPAs, overall GPA on UIUC courses are suggested to be above 2.5. In junior year, students

will apply for international transfer to be admitted to UIUC and change academic status from non-degree student to degree student in Fall semester of senior year.

To remain in good academic standing and to graduate from the ME curriculum, a student must have the curriculum courses satisfied with the TGPA or GPA requirements listed in section 8.

2.2 Graduation Requirements for ZJU degree

In addition to the 128-AU requirement listed above, the curriculum leading to the degree of Bachelor of Engineering from ZJU requires students to complete additional **33.5** credits of ZJU-required general education courses for domestic students and additional seven courses (**23** credits) in Chinese language and society study for international students. Please refer to section 5 for details.

3. Required Courses

3.1 Orientation and Professional Development

This course introduces students to the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. It also provides the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Academic Credit
ENG 100	Engineering Orientation	1
ME 290	Seminar	0
	Total Required AUs:	**1**

3.2 Foundational Mathematics and Science

These courses emphasize the fundamental mathematical and scientific principles upon which the engineering discipline is based.

Course Code[1]	Course Name	Academic Credits
MATH 221	Calculus I	4
MATH 231	Calculus II	3
MATH 241	Calculus III	4
MATH 257[2]	Linear Algebra with Computational Application	3
MATH 285	Intro Differential Equations	3
CHEM 102	General Chemistry I	3
CHEM 103	General Chemistry Lab I	1

PHYS 211	University Physics: Mechanics	4
PHYS 212	University Physics: Elec & Mag	4
	Total Required AUs:	**29**

[1] To register for third-year Mechanical Engineering (ME) courses, students are required to have a grade-point average (GPA) of 2.25 or above in the course except MATH 415.
[2] To remain in good academic standing and to graduate from the ME curriculum, a student must have a GPA of at least 2.0 in the course.

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of mechanical engineering.

Course Code	Course Name	Academic Credits
CS 101	Intro Computing: Engineering & Science	3
ECE 205[1]	Electrical and Electronic Circuits	3
ECE 206	Electrical and Electronic Circuits Lab	1
TAM 210[2]	Introduction to Statics	2
TAM 212	Introductory Dynamics	3
TAM 251	Introductory Solid Mechanics	3
ME 170	Computer-Aided Design	3
ME 270	Design for Manufacturability	3
ME 200	Thermodynamics	3
ME 310	Fundamentals of Fluid Dynamics	4
ME 320	Heat Transfer	4
ME 330	Engineering Materials	4
ME 340	Dynamics of Mechanical Systems	3.5
ME 360	Signal Processing	3.5
ME 370	Mechanical Design I	3
ME 371	Mechanical Design II	3
ME 470[3]	Senior Design Project	3
ME 498[3]	Senior Design Supplement	1

	Total Required AUs:	**53**

[1] ECE 205 can be substituted by ECE 110 (Introduction to Electronics, 3 AUs) + ECE 211 (Analog Circuit & System, 2 AUs) or ECE 210 (Analog Signal Processing, 4 AUs);
[2] TAM 210 can be substituted by TAM 211;
[3] ME 470 + ME 498 satisfy the Advanced Composition requirements.

3.4 Composition

i) These courses teach fundamentals of expository writing.

Course Code	Course Name	Academic Credits
RHET 101	Principles of Writing	4
RHET 102	Principles of Research	4
	Total Required AUs:	**8**

ii) The Advanced Composition requirement is fulfilled by a writing-intensive course beyond basic composition. It is normally taken in the junior or senior year.

Course Code	Course Name	Academic Credits
ME 470*	Senior Design Project	3
ME 498*	Senior Design Supplement	1
	Total Required AUs:	**4**

** ME 470+498 is also a required technical core.*

4. Elective Courses

4.1 Science Electives

The science electives augment the foundational science courses in an area of interest to the student and provide preparation for later courses. The science, statistics, and MechSE electives, and additional technical courses, stress the rigorous analysis, design, and statistics principles practiced in mechanical engineering.

Student must select one set of science electives from either Chem 104 and Chem 105, or Phys 213 and Phys 214, or MCB 150.

Course Code	Course Name	Academic Units
CHEM 104	General Chemistry II	3
CHEM 105	and General Chemistry Lab II	1
PHYS 213	Univ Physics: Thermal Physics	2
PHYS 214	and Univ Physics: Quantum Physics	2

MCB 150	Molecular & Cellular Basis of Life	4
	Total Required AUs:	**4**

4.2 Statistic Electives

Student must select **One** (**1**) course from the following list.

Course Code	Course Name	Academic Credits
IE 300	Analysis of Data	3
STAT 400/Math 463	Statistics and Probability I	4
ECE 313	Probability with Engineering Application	3
	Total Required AUs:	**3**

4.3 MechSE Electives

Students must take **Two (2)** courses from MechSE elective courses as listed in the following table.

Course Code	Course Name	Academic Credits
ME 400	Energy Conversion System	3
ME 401	Refrigeration and Cryogenics	3
ME 402	Design of Thermal Systems	3
ME 403	Internal Combustion Engines	3
ME 404	Internal Thermodynamics	4
ME 410	Intermediate Gas Dynamics	3
ME 411	Viscous Flow & Heat Transfer	4
ME 412	Numerical Thermo-Fluid Mechanics	3
ME 420	Intermediate Heat Transfer	4
ME 430	Failure of Engineering Materials	3
ME 431	Mechanical Component Failure	3
ME 432	Fundamentals of Photovoltaics	3
ME 440	Kinematics & Dynamics of Mechanical System	3
ME 445	Introduction to Robotics	4
ME 446	Robot Dynamics and Control	4
ME 447	Computational Design and Dynamics of Soft Systems	4

ME 451	Computer-Aided Manufacturing System	3
ME 452	Numerical Control of Manufacturing Processes	3
ME 453	Data Science in Manufacturing Quality Control	3
ME 455	Micromanufacturing Process & Automation	3
ME 458	Additive Manufacturing and Product Design	3
ME 460	Industrial Control System	4
ME 461	Computer Control of Mechanical System	3
ME 462	Advanced Computer Control	4
ME 471	Finite Element Analysis	3
ME 472	Introduction to Tribology	3
ME 475	Bioinspired Design	3
ME 481	Whole-Body Musculoskeletal Biomechanics	3
ME 482	Musculoskeletal Tissue Mechanics	3
ME 483	Mechanobiology	4
ME 485	MEMS Devices & Systems	3
ME 487	MEMS-NEMS Theory & Fabrication	4
ME 498VS	Vacuum Engineering in Semiconductor Industry	3
TAM 412	Intermediate Dynamics	4
TAM 413	Fund of Engineering Acoustics	3
TAM 416	Introduction to Nonlinear Dynamics and Vibrations	4
TAM 424	Mechanics of Structural Metals	3
TAM 428	Mechanics of Composites	3
TAM 435	Intermediate Fluid Mechanics	4
TAM 445	Continuum Mechanics	4
TAM 451	Intermediate Solid Mechanics	4
TAM 456	Experimental Stress Analysis	3
TAM 461	Cellular Biomechanics	4
TAM 470	Computational Mechanics	3

4.4 Technical Electives

Students must take **Two (2)** courses (6 AUs) from Technical elective courses as listed.

Agri. Bio Eng. (ABE): 361, 430*, 436, 445, 452*, 455, 456, 459, 466, 469, 476, 483, 488.

Aerospace Eng. (AE): 352, 402, 403, 410, 412, 416, 419, 420, 428, 433, 434, 442, 443, 451, 452, 454, 456, 460, 461, 482, 483.

Biochemistry (BIOC): 406, 446, 455

Bioengineering (BIOE): 380, 414, 415, 416, 461, 476, 479, 481, 482.

Biophysics (BIOP): 401, 419, 432.

Civil & Env. Eng. (CEE): 310, 330,340, 350, 360, 380, 401, 405, 406, 407, 408, 409, 410, 411, 412, 415, 416, 417*, 418, 420, 421, 422, 434, 437, 438, 440, 442, 443, 444, 447, 449, 450, 451, 453, 457, 458, 460,461, 462, 463, 465, 467, 468, 469, 470, 471, 472, 483, 484,491

Chem. & Biomol. Eng. (CHBE): 422,424,451,452,453,456,471,472,473,474,475,476.

Chemistry (CHEM): 232, 233, 236, 237, 312, 315, 317, 332, 420, 436, 437, 440, 442, 444, 445, 447, 460, 472, 474, 480, 482, 483, 488.

Computer Science (CS): 225, 233, 242, 341, 357, 374, 410, 411, 412, 413, 414, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 431, 433, 436, 438, 439 ,440, 445, 446, 447, 450, 460, 461, 463, 465, 466, 467, 468*, 473, 475, 476; 477, 481, 483, 484.

Computational Science and Eng. (CSE): 401, 402, 412, 441, 450, 451, 461.

Electrical & Computer Eng. (ECE): 329, 330, 333, 340, 342, 343, 380, 395, 401, 402, 403, 408, 411, 412, 414, 415, 416, 417, 418, 419, 420, 422, 424, 425, 428, 431, 432, 435, 437, 438, 439, 441, 444, 447, 448, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 462, 463, 464, 465, 466, 467, 468, 469, 470, 472, 473, 476, 478, 480, 481, 482, 483, 485, 486, 487, 488, 489, 490, 491, 492, 493, 495**.**

Economics (ECON): 302*.

Industrial Engineering (IE): 310, 311, 330, 340, 360, 410, 411, 412, 413, 420, 430, 431, 445*.

Mathematics (Math): 347, 357, 403, 412, 413, 414, 417, 418, 423, 424, 425, 427, 432, 442, 444, 446, 447, 448, 450, 453, 464, 473, 475, 481, 482, 484, 487, 489,

Molecular & Cellular Biology (MCB): 401, 402, 450, 493*.

Materials Science & Engineering (MSE): 304, 307, 308, 401, 402, 403, 405, 406, 420, 421, 422, 440, 441, 443, 450, 453, 455, 456, 457, 458, 460, 461, 466, 470, 473, 474, 480, 481, 485, 487, 488, 489.

Nuclear Plasma & Radiological (NPRE): 402, 412, 421, 423, 429, 435, 441, 442, 448, 451, 455, 457, 461, 470, 475.

Physics (PHYS): 330, 401, 402, 403, 404, 406, 427, 435, 436, 460, 466, 470, 475, 485, 486, 487, 496*

Systems Eng (SE): 400*, 402, 411, 412, 413, 420, 422, 423, 424, 450*.

Statistics (STAT): 409, 410, 420, 424, 425, 428, 429, 440, 448, 458, 480.

Technology Entrepreneurship (TE): 461*.

Technology and Management (TMGT): 460, 461

Mechanical Engineering (ME):** All of 400-level classes, except for 470, 497, 498.

Theoretical and Applied Mechanics (TAM):** All of 400-level classes, except for 497, 498

* This course falls under the Professional Elective category. No more than 3 hours of Professional Elective credit may be used to satisfy the Technical Elective requirements.

** The credits in excess of the 6 AU requirement of MechSE electives can fulfill Technical Elective requirements.

4.5 General Education

The general education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the General Education requirements, students must complete a minimum of **twelve (12)** AUs in general education courses. Students must take all courses for grade and complete sufficient courses in required categories based on the table below:

Required course category	Number of courses	Academic Credits
1) **Humanities & Arts**	Two (2)	6
2) **Social & Behavioral Sciences**[1]	Two (2)	6
Additional Requirements:		
i) Culture Studies[2]		
Western/Comparative Culture(s)	One (1)	3
Non-Western Culture(s)	One (1)	3
U.S. Minority Culture(s)[3]	One (1)	3
ii) Aesthetic Education[4]	One (1)	3
	Total required AUs:	**12**

[1] One of the Social and Behavioral Science courses must be an introductory economics course (ECON 102 or ECON 103).
[2] Proper selection of Social and Behavioral Sciences and in Humanities and the Arts courses will assure that these courses also fulfill Western, non-Western and US Minority Cultures requirements.
[3] Not required if students pursue ZJU degree only.
[4] Required only for ZJU degree.

For courses that fail in multiple categories of general education, the academic units of the course will be counted to satisfy the AU requirements of each category, but be counted only once in satisfying the total minimum academic unit requirements (12 AUs) of general education.

4.6 Free Electives

These unrestricted electives give students the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students define their research specialties. Students are encouraged to take cross-disciplinary courses as free electives.

Students must complete a minimum of **six (6)** AUs in free electives.

5. ZJU Required General Education

These courses introduce modern Chinese history, social development, government policies, etc., and help students improve their English and maintain a healthy lifestyle.

5.1 For domestic students

Domestic students must complete all the courses below to fulfill the graduation requirements for the Bachelor of Engineering Degree at ZJU. These courses can be taught in Chinese. The academic credits in this section are additional to 128 AUs of credits required for UIUC degree.

Course Code	Course Name	Academic Credits
LAW 1002	Ideology, Morality and Rule of Law	3
HIST 2001	Modern Chinese History	3
PHIL 2002	Introduction to the Principle of Marxism	3
PS 2011	Intro.to Mao Thought & Theoretical System of China Socialism	3
PS 3011	General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3
PS 1001	Situation and Policy I	1
PS 2001	Situation and Policy II	1
ENGL 1001	Integrated English I	1.5
ENGL 1002	Integrated English II	1.5
ENGL 2001	Advanced Spoken English I	1.5
ENGL 2002	Advanced Spoken English II	1.5
PE 1001	Physical Education I	1
PE 1002	Physical Education II	1
PE 2001	Physical Education III	1
PE 2002	Physical Education IV	1
PE 4021	Physical Education VII-Fitness test and exercise	0.5
MITR 1001	Military Training	2
MITR 2001	Military Theory	2

Course Code	Course Name	Academic Credits
ST 4001	Senior Thesis	2
	Total Required AUs:	**33.5**

5.2 For International students

International students are required to complete the following courses in Chinese language study and Chinese society to fulfill the graduation requirements along with the above 128 credits of courses towards the Bachelor of Engineering Degree at ZJU.

Course Code	Course Name	Academic Credits
CHIN 1001	Chinese I	4
CHIN 1002	Chinese II	4
CHIN 1003	Chinese III	4
CHIN 1004	Chinese IV	4
CHIN 1005	International Chinese Language Proficiency and Testing	2
CULT 2001	China Survey	3
ST 4001	Senior Thesis	2
	Total Required AUs:	**23**

6. Course Schedule Sample by Semester

6.1 First Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Rhet 101	Principles of Writing	4
2	Chem 102	General Chemistry I	3
3	Chem 103	General Chemistry Lab I	1
4	Math 221	Calculus I	4
5	ECE 110	Intro to Electronics	3
6	CS 101	Intro Computing: Engrg & Sci	3
7	Eng 100	Engineering Orientation	1
8	ME 290	Seminar	0

		Total:	19

List of ZJU courses for domestic students

No	Course Code	Course Name	Academic Credits
1	MITR 1001	Military Training	2
2	ENGL 1001	Integrated English I	1.5
3	PE 1001	Physical Education I	1
4	PS 1001	Situation and Policy I	

List of ZJU courses for International students

No	Course Code	Course Name	Academic Credits
1	CHIN1001	Chinese I	4

6.2 First Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Rhet 102	Principles of Research	4
2	Math 231	Calculus II	3
3	Phys 211	University Physics: Mechanics	4
4	Chem 104	General Chemistry II	3
5	Chem 105	General Chemistry II Lab	1
6	ME 170	Computer-Aided Design	3
		Total:	18

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	LAW 1002	Ideology, Morality and Rule of Law	3
2	ENGL 1002	Integrated English II	1.5
3	PE 1002	Physical Education II	1
4	PS 1001B	Situation and Policy I	1

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	CHIN 1002	Chinese II	4

6.3 Second Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Math 241	Calculus III	4
2	MATH 257	Linear Algebra with Computational Application	3
3	Phys 212	University Physics: Elec& Mag	4
4	TAM 211	Statics	3
5	ME 270	Design for Manufacturability	3
		Total:	17

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	PS 2011	Intro.to Mao Thought & Theoretical System of China Socialism	3
2	ENGL 2001	Advanced Spoken English I	1.5
3	PE 2001	Physical Education III	1
4	MITR 2001	Military Theory	2
5	PS 2001	Situation and Policy II	

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	CHIN1003	Chinese III	4
2	CULT2001	China Survey	3

6.4 Second Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Math 285	Intro to Differential Eq	3
2	TAM 212	Introductory Dynamics	3
3	TAM 251	Introductory Solid Mechanics	3
4	ME 200	Thermodynamics	3
5	ECE 211	Analog Circuits & Systems	2
6	GenEd	General Education Elective	3
		Total:	17

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	HIST 2001	Modern Chinese History	3
2	PHIL 2002	Introduction to the Principle of Marxism	3
3	ENGL 2002	Advanced Spoken English II	1.5
4	PS 2001	Situation and Policy II	
5	PE 2002	Physical Education IV	1

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	CHIN 1004	Chinese IV	4
2	CHIN 1005	International Chinese Language Proficiency and Testing	2

6.5 Third Year-First (Fall) Semester

No	Course Code	Course Name	Academic Credits
1	ME 340	Dynamics of Mechanical Systems	3.5
2	ME 330	Engineering Materials	4
3	ME 370	Mechanical Design I	3
4	STAT 400*	Statistics and Probability I	4
5	ECE 206	Elec & Electronic Circuits Lab	1
		Total:	15.5

**IE 300, Math 463, ECE 313 can be a substitute.*

6.6 Third Year- Second (Spring) Semester

No	Course Code	Course Name	Academic Credits
1	ME 310	Fundamentals of Fluid Dynamics	4
2	ME 371	Mechanical Design II	3
3	ME 360	Signal Processing	3.5
4	Free Elec	Free Elective	3
5	GenEd	General Education Elective	3
		Total:	16.5

6.7 Fourth Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	ME 320	Heat Transfer	4
2	MechSE E	MechSE Elective	3
3	MechSE E	MechSE Elective	3
4	Free Elec	Free Elective	3
5	GenEd	General Education Elective	3
		Total:	16

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	PS 3011	General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3
2	PS 2001	Situation and Policy II	

6.8 Fourth Year- Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	ME 470	Senior Design Project	4
2	ME 498	Senior Design Supplement	1
3	Tech Elec	Technical Elective	3
4	Tech Elec	Technical Elective	3
5	Free Elec	Free Elective	3
6	GenEd	General Education Elective	3
		Total:	17

List of ZJU Courses for domestic students

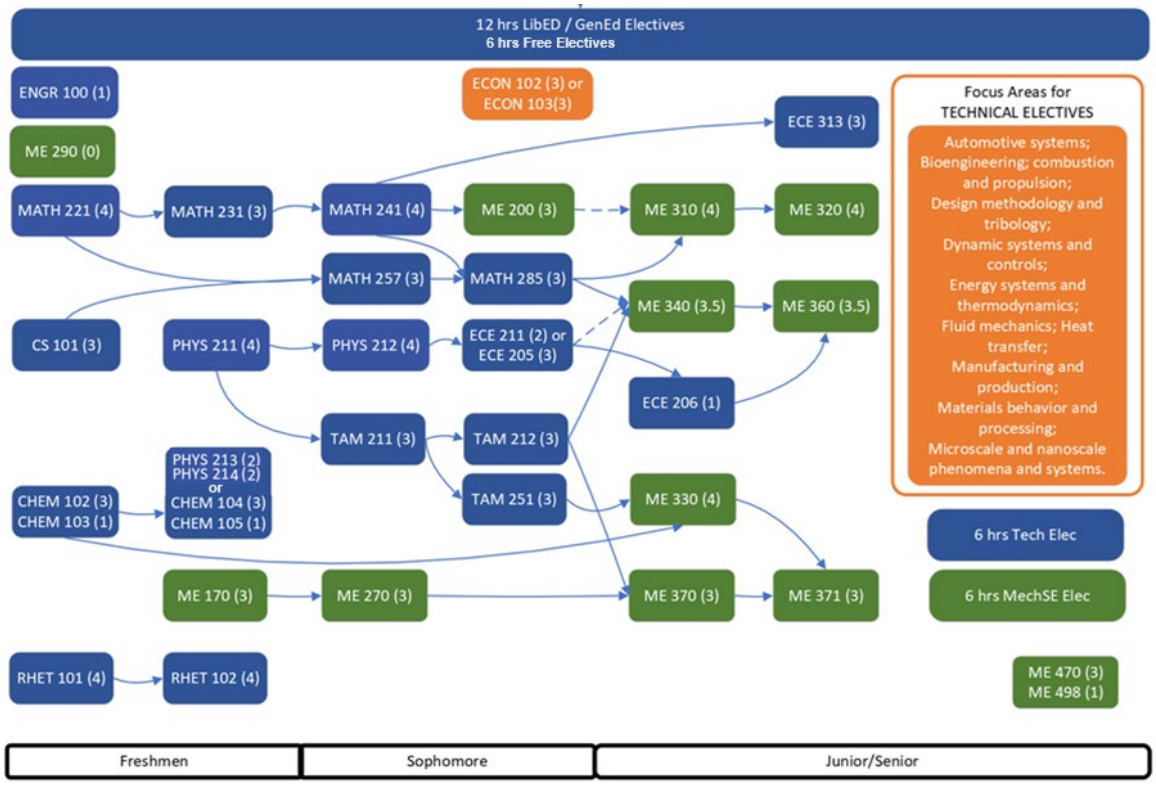
No	Course Code	Course Name	Academic Credits
1	PE 4021	Physical Education VII-Fitness test and exercise	0.5
2	PS 2001F	Situation and Policy II	1
3	ST 4001	Senior Thesis	2

List of ZJU Courses for international students

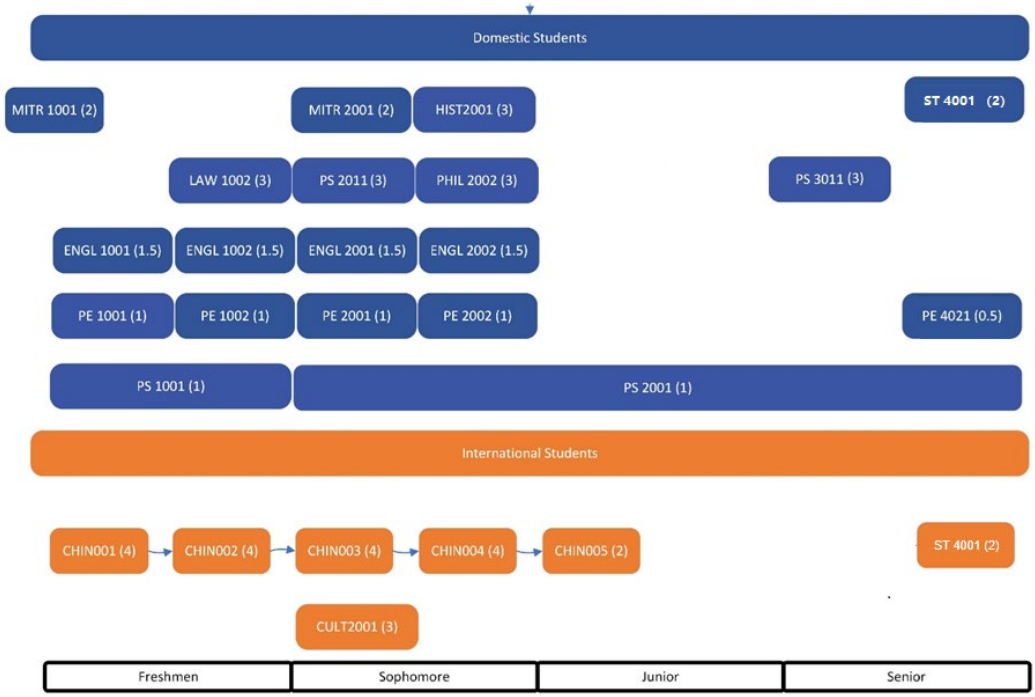
No	Course Code	Course Name	Academic Credits
1	ST 4001	Senior Thesis	2

7. Curriculum Flow Map

The following flow map provides a quick summary of the main features of the Mechanical Engineering curriculum.



ZJU Required General Education Course



8. Technical GPA (TGPA) and 2.25 GPA requirements on the Courses in Mechanical Engineering Curriculum

Course Rubric	Course Name	Credit	TGPA[1]	2.25 GPA[2]
Orientation and Professional Development				
ENG 100	Engineering Orientation	0	☐	☐
ME 290	Seminar	0	☐	☐
Foundational Mathematics and Science				
MATH 221	Calculus I	4	☐	■
MATH 231	Calculus II	3	☐	■
MATH 241	Calculus III	4	☐	■
MATH 257	Linear Algebra with Computational Application	3	■	☐
MATH 285	Intro Differential Equations	3	☐	■
CHEM 102	General Chemistry I	3	☐	■
CHEM 103	General Chemistry Lab I	1	☐	■
PHYS 211	University Physics: Mechanics	4	■	☐
PHYS 212	University Physics: Elec & Mag	4	■	☐
Mechanical Engineering Technical Core				
CS 101	Intro Computing: Engineering & Science	3	☐	■
ECE 205[3]	Electrical and Electronic Circuits	3	■	■
ECE 206	Electrical and Electronic Circuits Lab	1	■	■
TAM 210[4]	Introduction to Statics	2	■	■
TAM 212	Introductory Dynamics	3	■	■
TAM 251	Introductory Solid Mechanics	3	■	■
ME 170	Computer-Aided Design	3	☐	■
ME 270	Design for Manufacturability	3	■	☐
ME 200	Thermodynamics	3	■	☐
ME 310	Fundamentals of Fluid Dynamics	4	■	☐
ME 320	Heat Transfer	4	■	☐
ME 330	Engineering Materials	4	■	☐
ME 340	Dynamics of Mechanical Systems	3.5	■	☐
ME 360	Signal Processing	3.5	■	☐
ME 370	Mechanical Design I	3	■	☐
ME 371	Mechanical Design II	3	■	☐
ME 470	Senior Design Project	3	■	☐
ME 498	Senior Design Supplement	1	■	☐
Electives and Composition				
Rhet 101	Principles of Writing	4	☐	☐
Rhet 102	Principles of Research	4	☐	☐
Stat Elec	IE 300, STAT 400 / MATH 463	3	■	☐
Sci. Elec.	CHEM 104&105, MCB 150, PHYS 213&214	4	☐ ■	■ ■
Tech Elec	Chosen from list in 4.4, often in TGPA	6	■	☐

MechSE Elec	TAM and ME courses from Tech Elec list	6	■	☐
GenEd		12	☐	☐
Free Elec		6	☐	☐

[1] To remain in good academic standing and to graduate from the ME curriculum, a student must have a GPA of at least 2.00 in the courses marked with an "■".

[2] To register for third-year Mechanical Engineering (ME) courses, students are required to have a grade-point average of 2.25 or above in the courses marked with an "■".

[3] ECE 205 can be substituted by ECE 110 (Introduction to Electronics, 3 AUs) + ECE 211 (Analog Circuit & System, 2 AUs) or ECE 210 (Analog Signal Processing, 4 AUs);

[4] TAM 210 can be substituted by TAM 211;